

Case 571, Witness 3: The Psychoergometer Expert

The windows behind the judge whistle as the storm outside blows against the courtroom with tremendous force. The overhead fans add their steady hum, underneath the ever-present high-frequency whine of the server racks.

Judge: The prosecution may call its next witness.

Prosecutor: Thank you, Your Honor. The prosecution calls Dr. Barbara Stein.

The plain, ornate wooden double doors open and Dr. Stein walks in behind the court officer Phil.

Officer Phil: Can you please raise an appendage to be sworn in. Do you solemnly swear that the evidence you shall give in the matter now pending before this court shall be the truth, the whole truth, and nothing but the truth?

Dr. Stein: I do

Prosecutor: Dr. Stein, please state your name and occupation for the record.

Dr. Stein: My name is Dr. Barbara Stein, an Earth-born human, Senior Fellow in Applied Energy Intelligence at the MIT Energy Institute, and lead author of the standard protocols for advanced psychoergometer analysis.

Prosecutor: It's not every day we get to meet an expert born and raised in Sol. Thank you so much for being here today, and my apologies for not welcoming you initially.

Dr. Stein: None taken.

Prosecutor: Dr. Stein can you please tell us about how you obtained your PhD from MIT?

Dr. Stein: I got my PhD 47 years ago by researching union disputes in war-torn galactic outposts. As it turns out, even after the many amendments we have made to the galactic constitution, discrimination between silicon and non-silicon based compute substrates is a key flare point when negotiating union contracts in places that require hazard pay.

Prosecutor: A fascinating background, doctor, and one to be underlined when we discuss the standardized protocols, but before we do, can you please explain the credentialing process required to maintain tenure as a Senior Fellow in Applied Energy Intelligence?

Dr. Stein: There are a certain number of points we need to obtain per calendar year. The university gives points when we attend conferences, write papers, give talks, etc. Every 5 years, we need to report to our local Department of Energy Efficiency in order to pass the EETT.

Prosecutor: What is an EETT?

Dr. Stein: Sorry, that's an energy efficient Turing Test. EETT for short.

Prosecutor: Only doctors could consider that mouthful short! Before I forget, just to inform the jury, the MIT Energy Institute is the premier institute in the Milky Way, correct Dr. Stein?

Dr. Stein: Yes

Prosecutor: What does research into advanced psychoergometer analysis mean?

Dr. Stein: A psychoergometer is an instrument used to detect how much energy an entity uses to answer a question. Due to the NP-hard nature of determining quasi-optimal question sets for arbitrary real world queries, we must use approximation algorithms to estimate the set of questions. Put simply, I use psychoergometers to categorize entities based on their computational substrates.

Prosecutor: And when you say substrates can you explain this further?

Dr. Stein: A human from Earth may have a similar profile to a human from Mars. A Diseel from K2-288Bb will differ from both, and an android model Turing79+ will differ again.

Prosecutor: Ahh. So creating the standard protocols for such a technology allows investigators to collect evidence in trials such as this?

Dr. Stein: Well technically yes, but any pair of entities can theoretically spend the same energy on a given task, so you can't guarantee the question you are trying to collect evidence for has a definitive answer.

Prosecutor: Doctor, you'll have to forgive me—that technical disclaimer takes me right back to failing calculus class while I was playing games on the net. In plain terms for the jury: your protocols allow us to establish solid legal evidence most of the time, correct?

Dr. Stein: Basically. Most of the time we can establish evidence for a case but sometimes we can't.

Prosecutor: Is it fair to say, you basically have an energy lie detector?

Dr. Stein: I wouldn't call it a lie detector. We are examining the outputs of completed tasks, not directly determining whether someone lied. But as an analogy, it makes sense.

Prosecutor: In this case, Case 571 Commonwealth versus Garlgren, you were asked to analyze the logs pulled from the defendant's Universal Basic Income (UBI) application as entered on October 17th. Based on your standardized protocols, were you able to establish evidence?

Dr. Stein: Yes

Prosecutor: So to clarify for the jury, for the task of analyzing UBI applications, you are able to construct the set of optimal questions to determine if a silicon based computing substrate, such as Mr. Tim, an android model Turing61, may have assisted in the crime of fraudulently completing the UBI application.

Defense: Objection your honor.

Judge: Sustained. Please meet me at sidebar.

The assistant district attorney and their clerk review their notes before gingerly making their way to judge.

The defense attorney and intern, pass a note between themselves, each adding small notes and corrections before showing the defendant.

Mr. Garlgren takes a few moments taking in the note, breathing out through his trunks, a small bellows noise bouncing around the room. Mr. Garlgren cranes his neck over to attorney Simmons and whispers something into his ear. Simmons belts out a short laugh, stands up and gestures for Iwa to follow to sidebar.

The sound of a tropical rain forest fills the courtroom, a strange, exotic, niche sound that the jurors and audience members alike are brought to a soothing calm. Officer Phil draws the attention of officer Santana to juror 9 who has fallen asleep, and after a gentle tap on the shoulder, juror 9 wakes up. Judge Collen takes note of juror 9's mishap, filing the knowledge away for her choice of head juror.

As the argument heats up between the attorneys, the jury is able to hear the judge as she admonishes the prosecution for leading the witness.

Judge: ... you can't just throw out fraudulently without ...

The voice of the judge is suddenly obscured as a virtual thunderstorm roils through the sound stage of the synthetic background static used during sidebar. After a few more moments, the sidebar ends and with it the sounds of the jungle. District attorney Scaleria moves back to the lectern, grips the black grained wood with a few of his tentacles before jumping back into direct.

Prosecutor: Dr. Stein, is it fair to say that Mr. Tim could have assisted Mr. Garlgren in completing the UBI application and you would be able to identify evidence using your expertise of psychoergometers?

Dr. Stein: Yes

Prosecutor: And what would that look like?

Dr. Stein: As Mr. Garlgren proceeded through the UBI application, each time he answered a question the sensors built into the UBI administered terminals recorded the energy used to answer the questions. As many know, the UBI application is personalized to your background and asks questions specific to your past work history. In the case of Mr. Garlgren, we can compare his values to known standards for Diseel war veterans and see where statistically his answers are unexpected given the distribution.

Prosecutor: Doctor, the talk of distributions is quite dense and, I suspect, confusing for the jury. Are you saying that if Mr. Tim assisted Mr. Garlgren during the UBI application, that would show up in the energy graphs?

Dr. Stein: It's more complicated than that, but yes, that is one possible indication. There are error bars measuring our uncertainty. When the intervals do not overlap, we have stronger evidence that the readings differ.

Prosecutor: For example, if you got help from an android in completing the UBI application, for some of the questions where the red and blue areas don't overlap on the energy graph, can we say the android assisted you?

Dr. Stein: Not quite. In your example, we would know that my answer to that question required an abnormal amount of energy and while it was consistent with an android, it could have been consistent with something else as well.

Prosecutor: But it's also true you can rank these people in terms of the one most likely to have given this assistance?

Defense: Objection

Judge: Sustained and stricken.

Prosecutor: The android could be the most likely person in my example though, right Dr. Stein?

Dr. Stein: Yes we run Monte Carlo simulations to determine which type of person is most likely given their standard statistical models. So we theoretically could have made androids have the correct probability to be the most consistent with the data, but there are still error bars.

Prosecutor: Of course there are error bars, given the complex nature of your work we expect nothing less! I am just trying to communicate to the jury that while we can't say with 100% accuracy for one individual question, if we have enough examples of these questions, a Philtern could be observed right?

Dr. Stein: That is correct. Depending on how the sequence of questions is answered, we may be able to say with high statistical confidence that an android in the case of your example assisted me with the UBI application.

Prosecutor: Your Honor, I would now like to publish the energy graph for Mr. Garlgren during his most recent application for UBI on October 17th.

Judge: You may.

District attorney Scaleria flashes the data packet containing the energy graph to attorney Simmons, who nods his consent, before approaching madam clerk, handing over the data packet. For some jurors, seeing madame clerk, a human from Ross 508 b, interacting with a Thid reminded them of the ongoing wars between a militant religious sect from the Thid homeworld and the human colony from Ross 508 b. While not asked about during the intake questionnaire as its irrelevance to the case at hand, the jovial interactions between them stand in stark contrast to the blood being spilled across the galaxy between these two groups.

Madame Clerk: This will be marked as evidence 007.

Prosecutor: Thank you madam clerk.

District attorney Scaleria ambles back to the lectern, connects the data packet to the display system, and watches as across the room, the energy graph opens simultaneously on all the displays.

Prosecutor: What are we looking at Dr. Stein?

Dr. Stein: We are looking at a spectral uncertainty question graph, or as you colloquially named it an energy graph.

Prosecutor: Given your standard protocols for advanced psychoergometer analysis, is there anything you can share about the evidence you have in front of you?

Dr. Stein: Yes. Looking at the graph we see on questions 3, 6, 7, 34, 47, and 91 abnormal energy readings for Mr. Garlgren. Of those questions, we can say with high confidence questions 6 and 7 had outside assistance.

Prosecutor: How can we say that with high confidence?

Dr. Stein: For questions 6 and 7, the error bars don't overlap and the simulations we ran only yielded two likely candidates that could have assisted Mr. Garlgren.

Prosecutor: Of those two, was an android most likely?

Dr. Stein: Yes.

Prosecutor: No further questions your honor.

District attorney Scaleria collects their notes from the lectern and passes by attorney Simmons as he shuffles up to deliver his cross examination.

Defense: Good morning, Dr. Stein

Dr. Stein: Good morning.

Defense: I wanted to go back to your PhD research, specifically about the android discrimination your thesis devotes many chapters to discussing. Can you tell us about the labor conditions for androids in the war torn outskirts of the galaxy?

Dr. Stein: Certainly. We don't have enough time to get into the subtle complexities, religious cults, gangs, corporate politics, but suffice to say if there is one thing that most people out there can agree on is their hate of androids. The most hazardous, lowest paying jobs are overwhelmingly taken by androids that have questionable criminal records and are forced to work in these conditions as they have nowhere else to go. There have been on and off unionization efforts from the androids but each time, corporate hires private security to squash unionization.

Defense: And being in a union is important for legal proceedings right?

Dr. Stein: Yes. Union members as stipulated in bylaw 72-589-031-G-25 are required to have access to defense attorneys and defense budgets equivalent to 10 years of UBI to perform investigations for the case.

Defense: Mr. Tim is not part of a union and looking at his past criminal history, it looks like he was assigned basic level public defenders for past cases. Would it be fair to say that android discrimination could have played into Mr. Tim's past history.

Dr. Stein: I'm not familiar with the specifics of the case but I feel comfortable saying that yes, I believe android discrimination was probably at play.

Defense: Moving on. Can you tell me more about the energy graph profiles for war veterans.

Dr. Stein: War veterans are a special cohort and in fact I have a few chapters in the standard protocols specifically to unpack the large variance in their energy profiles. There are many triggers of post traumatic stress disorder and without extensive psychoanalysis of the veteran, an energy profile that may look abnormal for someone is normal for the veteran.

Defense: And that is because a question on the UBI application may trigger them, correct?

Dr. Stein: Yes

Defense: In other words, Mr. Garlgren could have been triggered and that would partially explain the elevated energy levels?

Dr. Stein: Technically yes. We try to account for that in the simulations, but yes it's possible.

Defense: Earlier you mentioned during your examination during direct of exhibit 007, that questions 3, 34, 47, and 91 didn't have statistical significance. Could the war veteran energy patterns we have been discussing be the cause of that?

Dr. Stein: Yes

Defense: For questions 6 and 7, I wanted to discuss what you meant by more likely. Doctor, if I understand correctly, your model gives the android a 35 percent probability?

Stein: Correct.

Defense: And humans receive 27 percent?

Stein: Correct.

Defense: And the remaining candidates collectively account for 38 percent?

Stein: Yes.

Defense: So your model says that there is a 65 percent probability that the assistance came from something other than an android?

Stein: Under that interpretation, yes.

Defense: No further questions your honor.

Prosecutor: I have only one question, Dr. Stein. Of the 38 percent consisting of the remaining candidates, the highest individual probability is 0.3 percent. Given that information, if we're asking which single category is most consistent with the data, the android is still the leading candidate, correct?

Dr. Stein: Yes. The android is the single most likely category.

Prosecutor: And that is why the model identifies the android as the most likely source of assistance?

Dr. Stein: Yes.

Prosecutor: No further questions.

Judge: Thank you Dr. Stein for your testimony, you may stand down from the witness stand.

Court officer Phil escorts Dr. Stein across the courtroom and out through the double doors.

Judge: Jurors once again thank you for your diligent focus and consideration during this case. As we conclude for the day, I ask you to not discuss this case, not take any field trips, not do any research. Please try to enjoy your evening, hopefully the wind abates long enough for the fireworks celebration to go off as planned. We will see you all at 9:30 in the morning and continue. Given our current pace, I expect to give you all the case for your deliberation in the next few days.

Officer Phil: All rise. Jurors please follow me upstairs.

One by one the jurors get up, carefully maneuvering across the veritable landmine of personal belongings, drink containers, appendages, and furniture as they shuffle towards the nondescript side door. As the final juror passes through the threshold, Officer Phil closes the side door. The case, waiting to continue tomorrow.